BD project(Final codes)

Spark_app.py:

---

mysql:

```
-- ==================== CREATE DATABASE ====================

CREATE DATABASE IF NOT EXISTS traffic_analysis;

USE traffic_analysis;


-- ==================== TABLE 1: Alcohol Risk Analysis (Narrow Table) ====================

CREATE TABLE alcohol_risk_analysis (

  id INT AUTO_INCREMENT PRIMARY KEY,

   region VARCHAR(100),

   time_of_day VARCHAR(50),

   driver_alcohol_level DECIMAL(5,3),

   accident_cause VARCHAR(200),

   alcohol_risk VARCHAR(20),

   batch_time TIMESTAMP,

   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

);

-- ==================== AGGREGATION TABLES (11 Tables) ====================


-- Table 2: Accidents by Region
```

```
CREATE TABLE accidents_by_region (
    id INT AUTO_INCREMENT PRIMARY KEY,
    region VARCHAR(100),
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    INDEX idx_region (region),
    UNIQUE KEY unique_region (region)
);

-- Table 3: Accidents by Urban/Rural
CREATE TABLE accidents_by_urban_rural (
    id INT AUTO_INCREMENT PRIMARY KEY,
    urban_rural VARCHAR(50),
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    INDEX idx_urban_rural (urban_rural),
    UNIQUE KEY unique_urban_rural (urban_rural)
);

-- Table 4: Accidents by Road Type
CREATE TABLE accidents_by_road_type (
    id INT AUTO_INCREMENT PRIMARY KEY,
    road_type VARCHAR(100),
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    INDEX idx_road_type (road_type),
```

```
    UNIQUE KEY unique_road_type (road_type)
);

-- Table 5: Accidents by Country
CREATE TABLE accidents_by_country (
    id INT AUTO_INCREMENT PRIMARY KEY,
    country VARCHAR(100),
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    INDEX idx_country (country),
    UNIQUE KEY unique_country (country)
);

-- Table 6: Top Accident Causes
CREATE TABLE top_accident_causes (
    id INT AUTO_INCREMENT PRIMARY KEY,
    accident_cause VARCHAR(200),
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    INDEX idx_accident_cause (accident_cause),
    UNIQUE KEY unique_accident_cause (accident_cause)
);

-- Table 7: Speed Limit Analysis
CREATE TABLE speed_limit_analysis (
    id INT AUTO_INCREMENT PRIMARY KEY,
    speed_limit VARCHAR(20),
```

```
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    INDEX idx_speed_limit (speed_limit),
    UNIQUE KEY unique_speed_limit (speed_limit)
);

-- Table 8: Driver Fatigue Analysis
CREATE TABLE driver_fatigue_analysis (
    id INT AUTO_INCREMENT PRIMARY KEY,
    driver_fatigue INT,
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    INDEX idx_driver_fatigue (driver_fatigue),
    UNIQUE KEY unique_driver_fatigue (driver_fatigue)
);

-- Table 9: Vehicle Condition Analysis
CREATE TABLE vehicle_condition_analysis (
    id INT AUTO_INCREMENT PRIMARY KEY,
    vehicle_condition VARCHAR(100),
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    INDEX idx_vehicle_condition (vehicle_condition),
    UNIQUE KEY unique_vehicle_condition (vehicle_condition)
);
```

```
-- Table 10: Road Condition Analysis
CREATE TABLE road_condition_analysis (
    id INT AUTO_INCREMENT PRIMARY KEY,
    road_condition VARCHAR(100),
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    INDEX idx_road_condition (road_condition),
    UNIQUE KEY unique_road_condition (road_condition)
);

-- Table 11: Weather Conditions Analysis
CREATE TABLE weather_conditions_analysis (
    id INT AUTO_INCREMENT PRIMARY KEY,
    weather_condition VARCHAR(100),
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    INDEX idx_weather_condition (weather_condition),
    UNIQUE KEY unique_weather_condition (weather_condition)
);

-- Table 12: Vehicles Most Involved
CREATE TABLE vehicles_most_involved (
    id INT AUTO_INCREMENT PRIMARY KEY,
    vehicle_type VARCHAR(100),
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
```

```
    INDEX idx_vehicle_type (vehicle_type),
    UNIQUE KEY unique_vehicle_type (vehicle_type)
);
```

```
SELECT 'alcohol_risk_analysis' AS table_name, COUNT(*) AS row_count FROM alcohol_risk_analysis
UNION ALL
SELECT 'accidents_by_region', COUNT(*) FROM accidents_by_region
UNION ALL
SELECT 'accidents_by_urban_rural', COUNT(*) FROM accidents_by_urban_rural
UNION ALL
SELECT 'accidents_by_road_type', COUNT(*) FROM accidents_by_road_type
UNION ALL
SELECT 'accidents_by_country', COUNT(*) FROM accidents_by_country
UNION ALL
SELECT 'top_accident_causes', COUNT(*) FROM top_accident_causes
UNION ALL
SELECT 'speed_limit_analysis', COUNT(*) FROM speed_limit_analysis
UNION ALL
SELECT 'driver_fatigue_analysis', COUNT(*) FROM driver_fatigue_analysis
UNION ALL
SELECT 'vehicle_condition_analysis', COUNT(*) FROM vehicle_condition_analysis
UNION ALL
SELECT 'road_condition_analysis', COUNT(*) FROM road_condition_analysis
UNION ALL
SELECT 'weather_conditions_analysis', COUNT(*) FROM weather_conditions_analysis
UNION ALL
SELECT 'vehicles_most_involved', COUNT(*) FROM vehicles_most_involved;
```

```
DROP DATABASE IF EXISTS traffic_analysis;
```

kafka-configs.bat --bootstrap-server localhost:9092 --alter --entity-type topics --entity-name test4 --add-config retention.ms=0

kafka-configs.bat --bootstrap-server localhost:9092 --alter --entity-type topics --entity-name test4 --add-config retention.ms=86400000:

```
-- Add ORDER BY to organize results better

SELECT 'alcohol_risk_analysis' AS table_name, COUNT(*) AS row_count FROM alcohol_risk_analysis

UNION ALL

SELECT 'accidents_by_region', COUNT(*) FROM accidents_by_region

UNION ALL

SELECT 'accidents_by_urban_rural', COUNT(*) FROM accidents_by_urban_rural

UNION ALL

SELECT 'accidents_by_road_type', COUNT(*) FROM accidents_by_road_type

UNION ALL

SELECT 'accidents_by_country', COUNT(*) FROM accidents_by_country

UNION ALL

SELECT 'top_accident_causes', COUNT(*) FROM top_accident_causes

UNION ALL

SELECT 'speed_limit_analysis', COUNT(*) FROM speed_limit_analysis

UNION ALL

SELECT 'driver_fatigue_analysis', COUNT(*) FROM driver_fatigue_analysis

UNION ALL
```

```
SELECT 'vehicle_condition_analysis', COUNT(*) FROM vehicle_condition_analysis

UNION ALL

SELECT 'road_condition_analysis', COUNT(*) FROM road_condition_analysis

UNION ALL

SELECT 'weather_conditions_analysis', COUNT(*) FROM weather_conditions_analysis

UNION ALL

SELECT 'vehicles_most_involved', COUNT(*) FROM vehicles_most_involved

-- Don't forget the raw_kafka_data table

UNION ALL

SELECT 'raw_kafka_data', COUNT(*) FROM raw_kafka_data

ORDER BY table_name;
```

```
SELECT * FROM driver_fatigue_analysis;
```

```
TRUNCATE TABLE alcohol_risk_analysis;

TRUNCATE TABLE accidents_by_region;

TRUNCATE TABLE accidents_by_urban_rural;

TRUNCATE TABLE accidents_by_road_type;

TRUNCATE TABLE accidents_by_country;

TRUNCATE TABLE top_accident_causes;

TRUNCATE TABLE speed_limit_analysis;

TRUNCATE TABLE driver_fatigue_analysis;

TRUNCATE TABLE vehicle_condition_analysis;

TRUNCATE TABLE road_condition_analysis;
```

```
TRUNCATE TABLE weather_conditions_analysis;

TRUNCATE TABLE vehicles_most_involved;



CREATE TABLE alcohol_risk_analysis (

    id INT PRIMARY KEY,  -- Kafka accident ID

    region VARCHAR(100),

    time_of_day VARCHAR(50),

    driver_alcohol_level DECIMAL(5,3),

    accident_cause VARCHAR(200),

    alcohol_risk VARCHAR(20),

    batch_time TIMESTAMP,

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

);



CREATE TABLE accidents_by_region (

    region VARCHAR(100) PRIMARY KEY, -- Changed: 'region' is now the primary key

    accident_count INT,

    batch_time TIMESTAMP,

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

    -- Removed 'id' column

);


-- Table 3: Accidents by Urban/Rural

CREATE TABLE accidents_by_urban_rural (

    urban_rural VARCHAR(50) PRIMARY KEY, -- Changed: 'urban_rural' is now the primary key

    accident_count INT,

    batch_time TIMESTAMP,
```

```
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -- Removed 'id' column
);

-- Table 4: Accidents by Road Type
CREATE TABLE accidents_by_road_type (
    road_type VARCHAR(100) PRIMARY KEY, -- Changed: 'road_type' is now the primary key
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -- Removed 'id' column
);

-- Table 5: Accidents by Country
CREATE TABLE accidents_by_country (
    country VARCHAR(100) PRIMARY KEY, -- Changed: 'country' is now the primary key
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -- Removed 'id' column
);

-- Table 6: Top Accident Causes
CREATE TABLE top_accident_causes (
    accident_cause VARCHAR(200) PRIMARY KEY, -- Changed: 'accident_cause' is now the primary key
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -- Removed 'id' column
```

```
);

-- Table 7: Speed Limit Analysis
CREATE TABLE speed_limit_analysis (
    speed_limit VARCHAR(20) PRIMARY KEY, -- Changed: 'speed_limit' is now the primary key
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -- Removed 'id' column
);

-- Table 8: Driver Fatigue Analysis
CREATE TABLE driver_fatigue_analysis (
    driver_fatigue INT PRIMARY KEY, -- Changed: 'driver_fatigue' is now the primary key
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -- Removed 'id' column
);

-- Table 9: Vehicle Condition Analysis
CREATE TABLE vehicle_condition_analysis (
    vehicle_condition VARCHAR(100) PRIMARY KEY, -- Changed: 'vehicle_condition' is now the primary key
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -- Removed 'id' column
);
```

```
-- Table 10: Road Condition Analysis
CREATE TABLE road_condition_analysis (
    road_condition VARCHAR(100) PRIMARY KEY, -- Changed: 'road_condition' is now the primary key
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -- Removed 'id' column
);

-- Table 11: Weather Conditions Analysis
CREATE TABLE weather_conditions_analysis (
    weather_condition VARCHAR(100) PRIMARY KEY, -- Changed: 'weather_condition' is now the primary key
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -- Removed 'id' column
);

-- Table 12: Vehicles Most Involved
CREATE TABLE vehicles_most_involved (
    vehicle_type VARCHAR(100) PRIMARY KEY, -- Changed: 'vehicle_type' is now the primary key
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -- Removed 'id' column
);
```

```
-- ==================== TABLE 1: Alcohol Risk Analysis ====================
```

```
CREATE TABLE alcohol_risk_analysis (
    id INT AUTO_INCREMENT,
    region VARCHAR(100),
    time_of_day VARCHAR(50),
    driver_alcohol_level DECIMAL(5,3),
    accident_cause VARCHAR(200),
    alcohol_risk VARCHAR(20),
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (id, region, time_of_day, driver_alcohol_level, accident_cause)
);

-- ==================== TABLE 2: Accidents by Region ====================
CREATE TABLE accidents_by_region (
    id INT AUTO_INCREMENT,
    region VARCHAR(100),
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (id, region)  -- Composite primary key
);

-- ==================== TABLE 3: Accidents by Urban/Rural ====================
CREATE TABLE accidents_by_urban_rural (
    id INT AUTO_INCREMENT,
    urban_rural VARCHAR(50),
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
```

```
    PRIMARY KEY (id, urban_rural)  -- Composite primary key
);

-- ==================== TABLE 4: Accidents by Road Type ====================
CREATE TABLE accidents_by_road_type (
    id INT AUTO_INCREMENT,
    road_type VARCHAR(100),
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (id, road_type)  -- Composite primary key
);

-- ==================== TABLE 5: Accidents by Country ====================
CREATE TABLE accidents_by_country (
    id INT AUTO_INCREMENT,
    country VARCHAR(100),
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (id, country)  -- Composite primary key
);

-- ==================== TABLE 6: Top Accident Causes ====================
CREATE TABLE top_accident_causes (
    id INT AUTO_INCREMENT,
    accident_cause VARCHAR(200),
    accident_count INT,
    batch_time TIMESTAMP,
```

```
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (id, accident_cause)  -- Composite primary key
);

-- ==================== TABLE 7: Speed Limit Analysis ====================
CREATE TABLE speed_limit_analysis (
    id INT AUTO_INCREMENT,
    speed_limit VARCHAR(20),
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (id, speed_limit)  -- Composite primary key
);

-- ==================== TABLE 8: Driver Fatigue Analysis ====================
CREATE TABLE driver_fatigue_analysis (
    id INT AUTO_INCREMENT,
    driver_fatigue INT,
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (id, driver_fatigue)  -- Composite primary key
);

-- ==================== TABLE 9: Vehicle Condition Analysis ====================
CREATE TABLE vehicle_condition_analysis (
    id INT AUTO_INCREMENT,
    vehicle_condition VARCHAR(100),
    accident_count INT,
```

```
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (id, vehicle_condition)  -- Composite primary key
);

-- ==================== TABLE 10: Road Condition Analysis ====================
CREATE TABLE road_condition_analysis (
    id INT AUTO_INCREMENT,
    road_condition VARCHAR(100),
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (id, road_condition)  -- Composite primary key
);

-- ==================== TABLE 11: Weather Conditions Analysis ====================
CREATE TABLE weather_conditions_analysis (
    id INT AUTO_INCREMENT,
    weather_condition VARCHAR(100),
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (id, weather_condition)  -- Composite primary key
);

-- ==================== TABLE 12: Vehicles Most Involved ====================
CREATE TABLE vehicles_most_involved (
    id INT AUTO_INCREMENT,
    vehicle_type VARCHAR(100),
```

```
    accident_count INT,
    batch_time TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (id, vehicle_type)  -- Composite primary key
);
```

Final schema:

```
CREATE DATABASE IF NOT EXISTS traffic_analysis;
USE traffic_analysis;

CREATE TABLE IF NOT EXISTS alcohol_risk_analysis (
    id INT PRIMARY KEY,
    region VARCHAR(100) DEFAULT 'Unknown',
    time_of_day VARCHAR(50) DEFAULT 'Unknown',
    driver_alcohol_level DOUBLE DEFAULT 0.0,
    accident_cause VARCHAR(255) DEFAULT 'Unknown',
    alcohol_risk VARCHAR(20),
    batch_time DATETIME
);

CREATE TABLE IF NOT EXISTS accidents_by_region (
    region VARCHAR(100) PRIMARY KEY,
    accident_count INT,
    batch_time DATETIME
);

CREATE TABLE IF NOT EXISTS accidents_by_urban_rural (
    urban_rural VARCHAR(50) PRIMARY KEY,
    accident_count INT,
```

```
    batch_time DATETIME
);
CREATE TABLE IF NOT EXISTS accidents_by_road_type (
    road_type VARCHAR(50) PRIMARY KEY,
    accident_count INT,
    batch_time DATETIME
);
CREATE TABLE IF NOT EXISTS accidents_by_country (
    country VARCHAR(100) PRIMARY KEY,
    accident_count INT,
    batch_time DATETIME
);
CREATE TABLE IF NOT EXISTS top_accident_causes (
    accident_cause VARCHAR(255) PRIMARY KEY,
    accident_count INT,
    batch_time DATETIME
);
CREATE TABLE IF NOT EXISTS speed_limit_analysis (
    speed_limit VARCHAR(50) PRIMARY KEY,
    accident_count INT,
    batch_time DATETIME
);
CREATE TABLE IF NOT EXISTS driver_fatigue_analysis (
    driver_fatigue INT PRIMARY KEY,
    accident_count INT,
    batch_time DATETIME
);
CREATE TABLE IF NOT EXISTS vehicle_condition_analysis (
    vehicle_condition VARCHAR(100) PRIMARY KEY,
```

```
    accident_count INT,
    batch_time DATETIME
);
CREATE TABLE IF NOT EXISTS road_condition_analysis (
    road_condition VARCHAR(100) PRIMARY KEY,
    accident_count INT,
    batch_time DATETIME
);


CREATE TABLE IF NOT EXISTS weather_conditions_analysis (
    weather_condition VARCHAR(100) PRIMARY KEY,
    accident_count INT,
    batch_time DATETIME
);
CREATE TABLE IF NOT EXISTS vehicles_most_involved (
    vehicle_type VARCHAR(100) PRIMARY KEY,
    accident_count INT,
    batch_time DATETIME
);
```